Exp 32:Implement a basic linear regression algorithm for predicting a continuous variable.

Code:

```
% Training data: Study Hours and Marks
data([1-2, 2-3, 3-5, 4-4, 5-6]).

% Calculate required sums
calculate([], 0, 0, 0, 0).

calculate([X-Y|T], SX, SY, SXY, SXX) :-
    calculate(T, A, B, C, D),
    SX is X + A,
    SY is Y + B,
    SXY is X*Y + C,
    SXX is X*X + D.

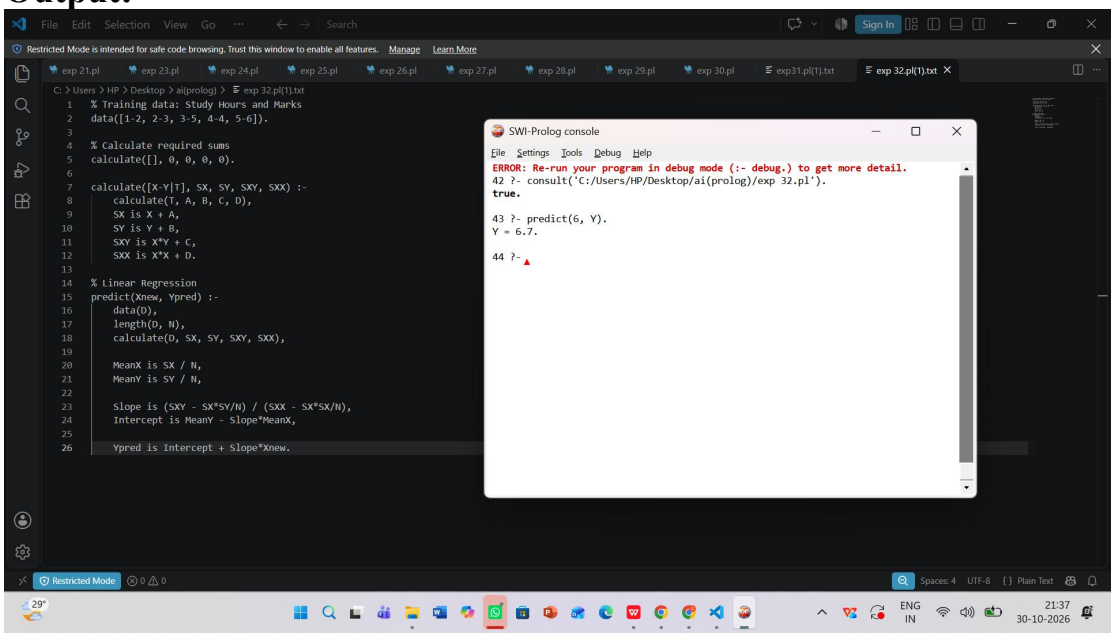
% Linear Regression
predict(Xnew, Ypred) :-
    data(D),
    length(D, N),
    calculate(D, SX, SY, SXY, SXX),

    MeanX is SX / N,
    MeanY is SY / N,

    Slope is (SXY - SX*SY/N) / (SXX - SX*SX/N),
    Intercept is MeanY - Slope*MeanX,

    Ypred is Intercept + Slope*Xnew.
```

Output:


% Training data: Study Hours and Marks
data([1-2, 2-3, 3-5, 4-4, 5-6]).
% Calculate required sums
calculate([], 0, 0, 0, 0).
calculate([X-Y|T], SX, SY, SXY, SXX) :-
calculate(T, A, B, C, D),
SX is X + A,
SY is Y + B,
SXY is X*Y + C,
SXX is X*X + D.
% Linear Regression
predict(Xnew, Ypred) :-
data(D),
length(D, N),
calculate(D, SX, SY, SXY, SXX),
MeanX is SX / N,
MeanY is SY / N,
Slope is (SXY - SX*SY/N) / (SXX - SX*SX/N),
Intercept is MeanY - Slope*MeanX,
Ypred is Intercept + Slope*Xnew.
SWI-Prolog console
File Settings Tools Debug Help
ERROR: Re-run your program in debug mode (:- debug.) to get more detail.
42 ?- consult('C:/Users/HP/Desktop/ai(prolog)/exp 32.pl').
true.
43 ?- predict(6, Y).
Y = 6.7.
44 ?-